LEARNING INSTITUTE of EMPOWERMENT (LIfE)

ACADEMIC ENGLISH (LAE1113)

TRIMESTER MARCH/APRIL (T2610)

FACULTY OF COMPUTING AND INFORMATICS (FCI)

SHOULD THE USE OF SINGLE-USE PLASTIC BAGS BE BANNED WORLDWIDE TO COMBAT ENVIRONMENTAL DEGRADATION

GROUP: 2

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FC02

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In this generation, the use of plastic materials is considered a norm in our daily lives. They can be seen in various forms of daily products, including food containers, bottles, cutleries and single-use plastic bags. Among these, single-use plastic bags are one of the most used items which can contribute to the environmental pollution. It is undoubtedly important for us to revise our lifestyle and consider using other alternatives methods or products to reduce the usage of plastic materials in our daily lives. As a result, there has been a global movement to ban on single-use plastic bags to protect marine life, reduce ongoing environmental and safeguard humans' health.

First and foremost, many believe that the widespread use of single-use plastic bags should be prohibited globally to curb the harm inflicted on marine biodiversity. This is largely because marine animals who often mistake plastic debris for food, leading to the ingestion of plastics. Due to their size, shape, and colour, which often resemble plankton species (Ding et al., 2018), microplastics are frequently ingested by marine invertebrates and fish. (Ayassamy, 2025, p. 2). This phenomenon can cause fatal digestive blockages for the marine animals. In addition, massive plastic waste causes accumulation of trash on coastlines, which leads to the fatal entanglement of marine life. This increases the risk of marine animals becoming trapped in the waste when they accidentally swim into it, making them unable to free themselves. For instance, seals may become tangled in plastic bands, causing them to struggle to breathe and eventually die. Not only that, but plastic bags also eventually break down into microplastics through exposure to sunlight, waves, and weathering. These microplastics then spread throughout the ocean and are ingested by small marine organisms. Larger marine animals then consume the contaminated small organisms, disrupting the marine ecosystem as the microplastics move up the marine food chain while carrying toxic chemicals. Thus, the global ban of single-use plastic bags is vital to protect marine biodiversity, as it would help preserve the stability of marine ecosystems for future generations.

Furthermore, the significant reason to prohibit plastic bags is their destructive impact on environment. Plastic bags contain chemical additives with varying level of toxicity, which can harm both human health and the environment. These substances have the potential to pollute the ground, air and water. Moreover, such pollutions occur through both the decomposition and the manufacturing of plastic bags. In terms of decomposition, plastic bags do not degrade naturally; instead, they take approximately around 20 to 500 years or more to degrade (Mentis et al., 2022, p. 2). Due to this reason, the accumulation of decomposing plastic bags in land and marine ecosystems leads to soil and water contamination, while their breakdown also emits harmful gases such as carbon dioxide and methane (Bowen Li et al., 2022, p. 4). Not only that, the manufacturing of the plastic bags also adds to serious environmental harm. This is because, the production of plastic bags relies on non-renewable natural resources such as natural gases and crude oil. The extraction process of these raw materials releases significant amounts of air pollutants into the atmosphere (Bowen Li et al., 2022, p. 1). Therefore, the combined effect of these processes is the release of vast amounts of toxic gases during plastic manufacturing and decomposition, which contribute to air pollution and add to the concentration of greenhouse gases, thus worsening global warming and climate change.

In addition to that, the use of plastic bags should be outlawed to ensure that humans' health is always protected from chronic diseases. This is because an excessive use of plastic bags will increase the risk of being exposed to the toxic materials in the plastic bags. As we all know, plastic bags are made of petroleum, which is a type of materials that contains heavy metal particles. These particles are often released in a significant amount when plastic bags are exposed to sunlight for a period of time (Karak, Parveen, Modak, Adhikari, & Chakrabortty, 2025, pp. 11 - 12). These released particles will be transmitted into food sources through grocery plastic bags or into water supplies through the breaking down of plastic bags in the soils, which will then be absorbed into our bodies. This action could cause serious health issues such as irritations, inflammations or even damage to the internal organs due to high concentration of heavy metals in our bodies (Karak et al., 2025, p. 12). This happens because heavy metals, such as Zinc and Lead, can carry microorganisms and malicious pathogens from contaminated surfaces into our organs (Karak et al., 2025, p.12). This as a result will increase the risk of being infected by deadly diseases. For example, cancer, cardiovascular problems and respiratory issues. Therefore, to avoid these chronic diseases, we should reduce the use of plastic bags in our daily life as it will decrease the chances of being in contact with toxic particles.

Next, plastic bags also provide several benefits in human life, as they offer many advantages and are convenient for everyday use (Li et al., 2022, p. 1). When it comes to grocery shopping, people usually require bags that are practical and easy to carry because groceries often include heavy items. Plastic bags are lightweight yet capable of carrying heavy loads. Furthermore, the use of plastic bags in the food industry is very widespread because plastic is a durable and airtight material. This helps to maintain the freshness of food for a longer period of time. However, the use of plastic bags can still cause a great deal of damage in the long term. For example, restaurants often use plastic materials such as polystyrene to wrap or package food. This is harmful because when plastic is exposed to high temperatures, its chemical structure may become unstable. As a result, toxic chemicals and microplastics from the polystyrene may seep into the food. Consequently, this may lead to long-term health risks such as cancer and other health problems (Li et al., 2022, p. 2). In addition, plastic bags that are not properly managed can negatively affect animals. For example, plastic bags that are thrown into the water may be mistaken by turtles for jellyfish. When turtles eat the plastic, it creates a false sense of fullness. As a result, they may stop eating real food and eventually die from starvation (Li et al., 2022, pp. 3 - 4). Therefore, the use of plastic bags not only affects human health but also harms animals and the environment.

In conclusion, it is essential to ban every single use of plastic bags globally to mitigate further harm to ecology. It is proven that plastic bags bring more negative consequences than benefits. If the use of plastic bags continues to grow over time, it will it will increase pollution and lead to the destruction of natural habits and biodiversity. For the future of our environment, urgent initiatives must be taken to prevent this crisis by all parties such as government and communities. Therefore, it is strongly hoped that the use of plastic bags can be reduced so that the beauty of the environment can continue to be enjoyed by future generations.

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